# House Price Prediction

Internship Project – Week 1 Summary

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## Project Overview

The objective of this project was to develop a machine learning model capable of predicting house prices using residential property characteristics and to identify the factors that most strongly influence property value. The Housing Prices Dataset from Kaggle containing 545 residential properties was used for the analysis. Data preprocessing included handling categorical variables, checking for duplicates, encoding features, and preparing the dataset for machine learning.

Two regression models, Linear Regression and Random Forest Regressor, were trained and evaluated. Exploratory data analysis, correlation studies, and visualization techniques were used to understand the relationships between different housing features and property prices.

## Model Performance

| **Metric** | **Value** |
|---|---|
| MAE | 970,043 |
| RMSE | 1,324,507 |
| $R^2$ Score | 0.653 |

The Linear Regression model achieved the best overall performance and explained approximately 65.3% of the variation in house prices.

## Key Findings and Business Insights

The analysis revealed that **area (0.536)**, **bathrooms (0.518)**, **air conditioning (0.453)**, **stories (0.421)**, and **parking availability (0.384)** were the most influential factors affecting house prices. One interesting observation was that modern amenities and furnishing status had a significant impact on property value, sometimes contributing more than traditional features such as the number of bedrooms.

Visualizations such as the house price distribution, correlation heatmap, amenity impact analysis, and feature ranking charts provided deeper insights into buyer preferences and market trends. Houses equipped with multiple amenities consistently achieved higher market values.

Based on these findings, real estate businesses should focus on larger properties equipped with modern amenities and located in desirable residential areas. Data-driven pricing strategies supported by machine learning can improve valuation accuracy, identify high-value property features, and support informed investment decisions.

*This project demonstrates how machine learning and data analytics can support accurate property valuation and data-driven decision-making in the real estate industry.*